

Invasion of 32-Bit MCU in 8-Bit And 16-Bit World

With the advancement in code density and core size, coupled with improved silicon processes, 32-bit microcontroller units have become affordable. High performance at low price has made 32-bit MCUs increasingly popular among designers



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When you go to buy a computer, you first look for a high-performance microprocessor and a large memory space. But, do you require all of that performance and memory at that moment? Not really, most of us anticipate the need to run more powerful applications and have more data storage in the future. Similarly, buying criteria for embedded products today includes expandable memory, connectivity to other devices and upgradability.

A 32-bit microcontroller unit (MCU) operates at a very high operating frequency with low power requirement and low cost, thus catering to the needs of today's embedded devices.

Integrated peripherals, on-chip memories and power performance are some of the key factors that have led to the growth of the 32-bit MCU market.

"Couple of years ago, we replaced our 8-bit design of the smart card terminal and fingerprint detection with 32-bit design. Our move was supported by reduced price of 32-bit MCU. The major advantage we saw was reduction in overall cost of the system due to decrease in component

Integrated peripherals, on-chip memories and power performance are some of the key factors that influence the choice of microcontrollers

counts. 8-bit design almost used 18-20 peripheral chips, whereas 32-bit used only five to six chips. 32-bit offers the advantages of integrated peripherals like I/O lines, TCP/IP and USB along with high performance," shares a design engineer from R&D department of a well-known manufacturer of electronic readers.

It's time to go 32-bit

Introduction of low-cost and energy-efficient 32-bit MCUs by leading manufacturers has resulted in a change in the trend. "32-bit MCU is usually more powerful in terms of architecture and operating frequency," says Wee Seng, senior manager, Renesas Electronics. Thus, with growing demand of features and functionality in embedded systems, 32-bit MCUs have started replacing the traditionally used 8-bit and 16-bit MCUs.

"Traditionally, embedded system developers selected an MCU that met the lowest common denominator for performance, memory, etc suitable for their application and within their cost target," says Ganesh Moorthy, CEO, Microchip. The choice was thus restricted to low-cost, low-performance 8-bit and 16-bit MCUs.

"For the control and display panel on the instruments, equipment and white goods, there is a trend of having better user-friendly interface. Instead of using an 8-bit MCU to interface with usual mechanical pushbutton, LED or LCD, the 32-bit 50MHz MCU is used to directly drive the QVGA TFT panel with touchscreen feature. This provides more graphical information